PROJECT FINAL REPORT

Global Olympic Sports Performance Analysis (1992–2026)

Date: 15 March 2026	**Team ID:** PNT2026TMID092418
Project Title: Global Olympic Sports Performance Analysis	**Maximum Marks:** Final Submission

1. INTRODUCTION

1.1 Project Overview

The Olympic Games serve as a global platform where nations compete to showcase sporting excellence. However, analyzing factors contributing to Olympic success remains challenging due to the massive volume of historical data spanning 1992 to 2026 (60 countries and 50 sports categories). This project delivers an end-to-end Business Intelligence solution combining Snowflake Data Warehouse engineering and Tableau visual analytics to evaluate per-athlete medal yield, regional investment efficiency, and host nation advantage across modern Summer and Winter Olympic Games.

1.2 Purpose

The primary purpose of this study is to provide data-backed evidence for three key stakeholders:
• **National Olympic Committees (Ms. Priya):** Identify high-yield sport categories (e.g., Rowing at 6.916 medals/athlete, Shooting at 6.731) to optimize national athlete funding.
• **International Funding Agencies (Mr. Carlos):** Evaluate capital investment vs. medal outcomes to spot high-efficiency, underfunded regions (South America leading at 49.64%).
• **Broadcast Media Networks (William Robert):** Craft compelling data stories on host nation dominance (Japan: 105, USA: 100, China: 93) and 34-year Olympic medal trends (305 in 2000 to 363 in 2020).

2. IDEATION PHASE

2.1 Problem Statement

PS No.	Customer Persona	Problem Statement (I am... I'm trying to... But... Because... Which makes me feel...)
PS-1	**Ms. Priya (NOC Strategist)**	I am an NOC strategist trying to identify high-yield sports categories to invest in, but historical data across 60 countries and 50 sports is massive and fragmented, because standard tallies lack per-athlete yield metrics, which makes me feel anxious about misallocating sports funding.
PS-2	**Mr. Carlos (Funding Analyst)**	I am an international grant analyst trying to evaluate sports investment vs. medal ROI, but it is difficult to spot overperforming under-resourced nations, because traditional metrics focus on wealthy nation medal totals, which makes me feel frustrated by capital inefficiency.
PS-3	**William Robert (Data Journalist)**	I am a media journalist trying to craft data-backed host nation dominance stories, but synthesizing 34 years of multi-game trends is complex, because total medals fluctuate (305 in 2000 vs 363 in 2020) and host data is scattered, which makes me feel overwhelmed by data density.

2.2 Empathy Map Canvas

The empathy map captures user behaviors, thoughts, touchpoints, and feelings across our core personas:

Persona	SAYS & THINKS	DOES & FEELS
Ms. Priya (NOC)	• 'We need to know exact medal ROI per athlete.' • Thinks: Wants data to justify shifting budget to Rowing/Shooting.	• Does: Analyzes 8 years of historical data across 50 sports. • Feels: Pressure to deliver high medal counts with finite budgets.
Mr. Carlos (Grants)	• 'Investment doesn't always equal success.' • Thinks: South America leads efficiency at 49.64%.	• Does: Plots investment vs medals in bubble charts. • Feels: Inspired by underfunded overperforming nations.
William Robert (Media)	• 'Home advantage creates dramatic stories.' • Thinks: Needs clear trends (Japan 105 host medals).	• Does: Traces 1992–2026 medal trend lines. • Feels: Excited to simplify complex data into visual stories.

2.3 Brainstorming & Idea Prioritization

Grouped brainstorming ideas were rated on feasibility and importance, selecting top priorities:

Group	Final Selected Idea	Feasibility	Importance	Priority	Selected
Group 1	Target NOC Investment in High-Yield Sports (Rowing 6.92, Shooting 6.73)	High	High	**Priority 1**	**Yes**
Group 2	Global Investment vs. Medal Efficiency Analysis (South America 49.64%)	Medium	High	**Priority 2**	**Yes**
Group 3	Broadcast Media Storytelling on Host Advantage (Japan 105, USA 100)	High	Medium	**Priority 3**	**Yes**

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Actions & Touchpoints	Thoughts & Feelings	Ownership & Opportunities
1. Discovery	Searches raw CSV dataset across 50 sports. Touchpoint: Raw dataset files.	Thought: 'Which sports give highest ROI?' Feeling: Overwhelmed by raw data.	Owner: Data Eng. Opp: Automate cloud staging pipelines.
2. Analysis	Filters Tableau dashboard by country/sport. Touchpoint: Interactive BI Dashboards.	Thought: 'Rowing gives 6.916 medals/athlete!' Feeling: Empowered by visual trends.	Owner: BI Developer. Opp: Add predictive forecasting models.
3. Decision	Drafts NOC sports budget memo & grants. Touchpoint: Executive story presentations.	Thought: 'We have proof to reallocate funds.' Feeling: Accomplished & confident.	Owner: Executive Board. Opp: Create automated PDF export templates.

3.2 Solution Requirements

Functional Requirements (FR):

• FR-1: Access & merge 1992–2026 Olympic athlete demographic and medal datasets across 60 nations.
• FR-2: Clean, standardize, and impute missing age/category attributes in Snowflake staging.
• FR-3: Compute key measures: Per-athlete yield (Rowing 6.916), Regional Efficiency (South America 49.64%), Host Medals (Japan 105).
• FR-4: Deliver Tableau dashboards featuring horizontal bar, bubble, pie/donut, and multi-decade line charts.
• **Non-Functional Requirements (NFR):** Usability (intuitive legends), Security (RBAC in Snowflake), Reliability (99.9% data accuracy), Performance (sub-1.2s query scan), Scalability (supports future 2028/2032 Games).

3.3 Data Flow Diagram (DFD)

• **External Entities:** Ms. Priya (NOC Strategist), Mr. Carlos (Funding Analyst), William Robert (Data Journalist).
• **Processes:** 1.0 Data Ingestion & Validation -> 2.0 Calculate Per-Athlete Yield & Efficiency -> 3.0 Generate Tableau Analytics.
• **Data Stores:** DS-1 Athletes Store, DS-2 Medal Results Store, DS-3 Financial & Regional Investment Store.
• **Data Flows:** Raw Flat CSVs -> Clean Staging Records -> Aggregated KPI Ratios -> Interactive Visual Dashboards.

3.4 Technology Stack

Layer	Technology Used	Justification & Purpose
Query / Programming	Python 3.11, ANSI SQL	Python pandas/numpy for statistical modeling; SQL for fast database aggregations.
Data Warehouse	Snowflake Data Warehouse	Cloud storage for high-volume historical athlete records and multi-year medal tallies.
ETL / Transformation	Snowflake SQL, Tableau Prep	Cleans demographic records, handles NULLs, creates star schema fact/dimension tables.
Visualization / Front-End	Tableau Desktop, Streamlit	Delivers interactive dashboards, bubble charts, donut charts, and narrative story points.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The Problem-Solution Fit canvas maps user constraints to our BI analytical features. By moving from unstructured flat files to central Snowflake views (`VW_PER_ATHLETE_YIELD`), NOC strategists transition from uncertainty to confident, data-backed budget allocation.

4.2 Proposed Solution

Our proposed BI solution implements automated ETL pipelines and interactive dashboards delivering:
1. **Horizontal Bar Charts:** Ranking per-athlete medal returns by sport category.
2. **Investment vs Medal Bubble Charts:** Identifying overperforming nations relative to capital budgets.
3. **Regional Efficiency Pie Charts:** Demonstrating South America's 49.64% efficiency leadership.
4. **Host Advantage & Trend Line Charts:** Tracking host dominance (Japan 105) and 1992–2026 total medal trends.

4.3 Solution Architecture

Layer	Role in Architecture	Technology Used
1. Data Sources	Historical Olympic records (1992–2026), 60 nations, 50 sports.	Flat CSV files (`athletes.csv`, `medal_results.csv`)
2. Ingestion	Automated staging into cloud data warehouse.	Snowflake COPY INTO, Internal Stages
3. Processing	Deduplication (`ROW_NUMBER()`), median null imputation.	Snowflake SQL, Tableau Prep Builder
4. Storage	Star schema dimension and fact table repository.	`OLYMPIC_PERFORMANCE_DB` (MART Schema)
5. BI Analytics	KPI calculation (Yield, Efficiency %, Host Flag).	Tableau Calculated Fields, Snowflake Views
6. Visualization	Executive dashboards and narrative story points.	Tableau Desktop / Public, Streamlit
7. Business Users	Executive decision-makers interacting with reports.	NOC Strategists, Grant Directors, Media Executives

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was executed using Agile Scrum framework across 4 Sprints (80 Total Story Points, 20 Pts/Sprint velocity):

Sprint	Epic / Focus Area	Completed User Stories	Points	Release Date
Sprint 1	Data Integration & Yield	USN-1 to USN-5: Snowflake staging, per-athlete yield bar chart.	**20**	29 Oct 2026
Sprint 2	Investment & Efficiency	USN-6 to USN-9: Investment bubble chart, regional efficiency pie chart.	**20**	05 Nov 2026
Sprint 3	Host Dominance & Trends	USN-10 to USN-13: Host advantage chart, 1992–2026 line trend.	**20**	12 Nov 2026
Sprint 4	Polish, UAT & Release	USN-14 to USN-17: UI refinement, security RBAC, final production release.	**20**	19 Nov 2026

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test Parameter	Snowflake Engine Benchmark	Tableau BI Engine Benchmark
Data Rendered	38,574 joined rows across 8 tables (26.5 MB)	38,574 historical medal records loaded
Data Quality	Passed 100%. Imputed 2,410 NULLs; removed 1,240 dups.	Zero null values across core visual dimensions
Filter Efficiency	Unfiltered: 840ms -> Single Country Filter: 180ms	Interactive parameter switch response = 1.1s
Visual Elements	8 Streamlit analytical charts rendered	8 Tableau visual charts across 2 dashboards
Highest Execution Query	Fact Aggregation Join = 1.24s (1240ms)	Extract Cold Load = 420ms (sub-50ms cached)

7. RESULTS

7.1 Output Screenshots & Core Analytical Takeaways

The integrated Snowflake and Tableau solution produced five primary data-driven outcomes:
1. **Rowing and Shooting Lead Per-Athlete Medal Yield:** Rowing achieves **6.916 average medals per athlete**, and Shooting achieves **6.731**, proving that technical density sports deliver superior per-athlete ROI.
2. **Global Powerhouse Dominance:** The **United States leads overall cumulative totals with 735 medals**, followed by **China with 696 medals**.
3. **South America Leads Capital Efficiency:** Regional breakdowns confirm **South America leads efficiency at 49.64%**, followed by Europe/Asia at **45.98%**. Nations like Slovakia, Georgia, and Ethiopia overperform relative to lower budgets.
4. **Measurable Host Advantage Boost:** Home host dominance is led by **Japan (105 host medals)**, USA (100), and China (93), demonstrating a 25–40% performance bump on home soil.
5. **34-Year Olympic Expansion:** Total medals awarded expanded from **305 in Sydney 2000** to **363 in Tokyo 2020**, recording **334 in 2026** alongside near 50-50 gender parity.

8. ADVANTAGES & DISADVANTAGES

Advantages:
• Evidence-based NOC budget optimization targeting high-yield disciplines (Rowing/Shooting).
• Objective capital efficiency metrics guiding international grant allocations to underfunded regions.
• Sub-1.2 second dashboard query performance powered by Snowflake micro-partition pruning.
• Comprehensive coverage of 34 years of historical Olympic competition (1992–2026).

Disadvantages / Limitations:
• Batch CSV data ingestion requires manual triggers for newly concluded Olympic Games.
• Current scope focuses on descriptive historical analytics rather than predictive machine learning forecasts.

9. CONCLUSION

The Global Olympic Sports Performance Analysis project successfully bridges the gap between massive historical sports data and executive decision-making. By proving that athletic success is driven by strategic capital allocation in high-yield sports (Rowing/Shooting) and targeting funding to high-efficiency regions (South America), the solution empowers NOC strategists, grant analysts, and media networks with undeniable visual evidence.

10. FUTURE SCOPE

• **Phase 2 (Q3 2026):** Machine Learning Predictive Models forecasting 2028 Los Angeles & 2032 Brisbane Olympic medal tallies.
• **Phase 3 (Q4 2026):** Automated Real-Time Pipeline Integration with IOC and financial sports APIs.
• **Phase 4 (Q1 2027):** Mobile-Native Analytics App with automated executive PDF report generation.

11. APPENDIX

Source Code (Snowflake DDL/DML Sample)

```
-- Snowflake Mart Layer Fact Table DDL
CREATE OR REPLACE TABLE OLYMPIC_PERFORMANCE_DB.MART.FACT_MEDAL_ACHIEVEMENTS AS
SELECT
m.medal_id, m.athlete_id, a.athlete_name, a.gender,
m.olympic_year, m.game_season, m.sport_category, m.event_name, m.medal_type,
n.country_name, n.region, i.sports_investment_usd,
CASE WHEN m.country_code = h.host_country_code THEN 1 ELSE 0 END AS is_host_nation
FROM OLYMPIC_PERFORMANCE_DB.STAGING.STG_MEDAL_RESULTS m
JOIN OLYMPIC_PERFORMANCE_DB.STAGING.STG_ATHLETES a ON m.athlete_id = a.athlete_id
JOIN OLYMPIC_PERFORMANCE_DB.CURATED.DIM_NATIONS n ON m.country_code = n.country_code
LEFT JOIN OLYMPIC_PERFORMANCE_DB.RAW.RAW_INVESTMENT i ON m.country_code = i.country_code AND m.olympic_year = i.olympic_year
LEFT JOIN OLYMPIC_PERFORMANCE_DB.RAW.RAW_HOSTS h ON m.olympic_year = h.olympic_year;
```

Dataset Link & Repository Access

- **Dataset Link:** Kaggle Historical Olympic Dataset (1992–2026) & IOC Official Reports (`athletes.csv`, `medal_results.csv`)
- **GitHub Repository Link:** https://github.com/olympic-analytics/snowflake-olympic-performance-2026
- **Tableau Public Dashboard Link:** https://public.tableau.com/views/GlobalOlympicSportsPerformanceAnalysis_2026/Dashboard1
- **Tableau Public Story Link:** https://public.tableau.com/views/GlobalOlympicSportsPerformanceStory_2026/Story1